Signal-to-Rain

Turning mobile networks into flood early warning sensors

Product Document | Nokia Student Network Challenge 2026, AI Challenge
Prepared by Emmanuel Safo and team, Tampere University | 27 October 2026

1. Summary

Signal-to-Rain is an AI service that uses the microwave radio links between cell towers as rain sensors. Rain weakens these signals, and operators already log signal strength every few minutes. Signal-to-Rain converts that signal loss into live rainfall maps and neighbourhood-level flood warnings, with no new hardware. The first target market is Ghana, followed by other African countries where weather radars are scarce but mobile networks are dense.

> *"When rain falls between two cell towers, the signal gets weaker. We measure that loss across thousands of links and turn the whole network into a rain radar."*

2. The Need (Idea)

Floods are frequent and deadly across Africa, and most of the continent lacks the observation network needed for timely warnings.

- Ghana loses about USD 200 million a year to floods and droughts, according to the Ghana Meteorological Agency (GMet).
- The June 2015 Accra flood claimed more than 200 lives.
- UNDP projections suggest Ghana's annual flood losses could exceed USD 500 million by 2030.
- The US and EU combined operate 636 weather radars for 1.1 billion people; Africa has 37 for 1.2 billion (WMO).
- Around 60 percent of Africans are not covered by early warning systems (WMO).
- Over 80 percent of Africans have access to a mobile network, so the infrastructure for both sensing and alerting already exists.

3. The Product (What)

A cloud service and web application for meteorological agencies, disaster management organisations, cities, insurers and mobile operators.

Capability	Description
Live rainfall map	Minute-level rainfall estimates along every microwave link, interpolated into a city or national map.
Flood risk forecast	AI model combining rainfall with terrain and drainage data to rank neighbourhoods by flood risk for the next hours.
Alerts	Automatic SMS and dashboard alerts to residents, NADMO and city assemblies.
Data API	Rainfall time series for insurers, agriculture and researchers.
Validation view	Side-by-side comparison with rain gauges and radar, where available.

4. How It Works

Existing network data in, early warnings out. No new hardware.

- **Sensors already exist.** Cell towers are linked by point-to-point microwave radio, typically around 6 to 80 GHz. Operators log the received signal level of each link for network monitoring.
- **Rain attenuates the signal.** Raindrops absorb and scatter microwaves, so heavier rain along a link means a weaker received signal.
- **Wet or dry classification.** An AI classifier separates rain events from normal fluctuations caused by wind, temperature and multipath. This is the hardest step.
- **Attenuation to rain rate.** Rain-induced attenuation is the drop from the dry baseline, corrected for water on the antennas. Dividing by link length gives specific attenuation k (dB/km), converted to rain rate with the ITU-R P.838 power law, $R = (k/a)^{1/b}$, where a and b depend on frequency and polarisation.
- **Rainfall map.** Thousands of crossing links are combined by spatial interpolation into a rainfall field.
- **Flood prediction and alerts.** A second model uses rainfall, terrain and drainage to predict flooding and triggers alerts.

5. Technical Architecture and Stack

Layer	Technology	Purpose
Data processing	Python, pycomlink, xarray, NumPy, pandas, SciPy	Load link data, baselines, wet antenna correction, rain rate
AI models	scikit-learn or LightGBM (PyTorch optional)	Wet/dry classification, flood risk scoring
Backend API	FastAPI, SQLite or files	Serve links, rainfall maps, gauge comparisons, alerts
Web app	React, Vite, Leaflet or MapLibre, Recharts, Tailwind CSS	Interactive map, storm replay slider, charts, alert panel
Tooling	Git, GitHub, Docker Compose, Linux	Collaboration and reproducible run
Edge stretch goal	C++ core with pybind11	Rain rate computation suitable for running on link equipment

Demo data: OpenMRG, an open dataset from SMHI, Ericsson and Hi3G Sweden with commercial microwave link data, weather radar and rain gauges for Gothenburg. It allows the demo to prove that estimated rainfall matches real gauge measurements.

6. Hackathon Build Plan (7 hours)

Time	Work	Output
Hour 0 to 1	Set up repo and Docker; load OpenMRG; pick a storm event	Clean link and gauge data
Hour 1 to 3	Rain rate pipeline with pycomlink; train wet/dry classifier	Rain rate per link, accuracy vs gauges
Hour 3 to 5	FastAPI endpoints; React map with storm replay and charts	Working web app

Time	Work	Output
Hour 5 to 6	Flood risk scoring and simulated SMS alert panel; Ghana mock-up	End-to-end demo
Hour 6 to 7	Slides, business case, rehearsal	5-minute pitch

7. Market and Business Model (How Much)

Customers: national meteorological agencies (GMet), disaster management (NADMO), city assemblies, insurers, agribusinesses and mobile operators. Public buyers are often funded by the World Bank and climate adaptation funds.

Revenue stream	Model (assumption)
National subscription	EUR 150,000 to 300,000 per country per year, depending on coverage
Data API	Tiered subscription for insurers, agribusiness and researchers
SMS rainfall and flood services	Small per-user fee, sold and billed through the operator
Operator revenue share	Operators receive a share for providing link data

Costs: cloud processing and storage, a small data science and support team, and one-time integration with each operator's network management system. No sensors need to be installed.

Value argument: the WMO estimates that USD 800 million invested in early warning systems in developing countries would prevent USD 3 to 16 billion in annual losses. If Signal-to-Rain prevents just 1 percent of Ghana's yearly flood and drought losses, that is about USD 2 million saved, many times the subscription price.

Prices above are working assumptions for the pitch and have not been validated with customers.

8. Go to Market (To Whom)

- **Phase 1, Ghana pilot:** one operator (MTN Ghana or Telecel) and GMet in Greater Accra, the most flood-exposed region.
- **Phase 2, Ghana national:** full network coverage with NADMO integration and SMS alerts.
- **Phase 3, West Africa and beyond:** replicate with operator groups present in many African countries.
- **Channel:** sold through mobile operators as a value-added service, with Nokia as the technology partner already supplying network equipment.
- **Timing:** Ghana awarded 5G spectrum directly to Telecel and MTN in 2026 after ending the exclusive NGIC wholesale model. Both operators are now building new networks, which is the moment vendors compete on added value.

9. Competitive Landscape

Player	What exists	Our difference
Ericsson and SMHI (Sweden)	Semi-operational rainfall from microwave links in Gothenburg and Stockholm since 2015; marketed as Ericsson Weather Data	Focus on Africa, where radar gaps are largest, plus AI flood risk on top of rainfall
Research groups (Germany, Netherlands, Italy, Belgium)	Country-scale studies and open tools such as pycomlink and RAINLINK	Productised service with alerts and a business model
Weather radars	Accurate but expensive and scarce in Africa	Uses existing infrastructure; complements radar instead of replacing it

The science is proven in Europe. The opportunity is to make it a product for regions that need it most, and to give Nokia an answer to Ericsson's offering.

10. Why Nokia

- Nokia supplies microwave backhaul and radio equipment, so the data source is its own hardware.
- Nokia was the strategic technology partner of Ghana's NGIC 5G venture and built its 5G core, giving it existing relationships in the country.
- Signal-to-Rain turns network equipment into a public safety service, a differentiator when operators choose vendors.
- It fits the AI challenge themes of sustainability and future communications, with a resilience angle shared with the Defense challenge.

11. Risks and Mitigations

Risk	Mitigation
Operators reluctant to share network data	Process data inside the operator's systems and share only rainfall outputs; offer revenue share
Accuracy without local calibration	Calibrate with GMet gauges; use AI wet/dry models; combine with satellite rainfall
Network changes (links added or removed)	Automatic link metadata updates and quality flags
Flood prediction needs local terrain and drainage data	Start with known flood hotspots in Accra and improve with city data
Public sector sales cycles are slow	Pilot funded by development partners; operator channel for faster entry

12. Roadmap

Stage	Milestone
Hackathon	Working demo on OpenMRG data with web app and simulated alerts
Months 1 to 3	Partnership talks with one Ghanaian operator and GMet; historical data test
Months 4 to 9	Accra pilot with live data and NADMO alert integration
Months 10 to 18	National rollout in Ghana; second West African country
Later	Edge processing on link equipment; integration with operator SMS alert systems

13. Team Roles

Role	Responsibility
Signal processing and AI (Emmanuel)	Rain rate pipeline, wet/dry classifier, flood scoring
Backend	FastAPI service and data endpoints
Web app	Map, storm replay, charts and alert panel
Business and pitch	Market figures, Ghana mock-up, slides and rehearsal

14. Sources

- Ghana News Agency (2025). Ghana loses USD 200 million annually to disasters, NADMO. gna.org.gh

- NAN News (2026). Ghana loses USD 200m annually to floods, droughts, GMet. nannews.ng
- 3News. Ghana's endless floods (UNDP projection). 3news.com
- Yale Environment 360. A lack of weather stations is costing African lives (WMO radar figures). e360.yale.edu
- African Arguments. WMO early warning investment estimate. africanarguments.org
- University of Cambridge, Department of Engineering. Investment in Africa's weather stations. eng.cam.ac.uk
- Andersson et al. (2022). OpenMRG: open data from microwave links, radar and gauges. Earth System Science Data 14, 5411 to 5426.
- van de Beek et al. (2019). Rain monitoring with commercial microwave link networks in Sweden. KIT symposium.
- Ericsson (2019). Ericsson Weather Data, microwave induced rain monitoring. unfccc.int
- Graf et al. (2020). Rainfall estimation from a German-wide CML network. Hydrology and Earth System Sciences 24, 2931 to 2950.
- Roversi et al. (2020). Commercial microwave links for operational rainfall monitoring in Northern Italy. AMT 13, 5779.
- Overeem (2023). Commercial microwave link data for rainfall monitoring, KNMI dataset. data.4tu.nl
- Tech Africa News (2024). Nokia expands partnership with NGIC to build Ghana's first 5G core. techafricanews.com
- Mobile World Live and DatacenterDynamics (2026). Ghana awards 5G licences to Telecel and MTN.